

LC 3342

UBER

Find Minimum Time to Reach Last Room II

Problem Breakdown & Notes

Basically a 2D grid, where each cell represents the time after which the room opens, we need to go from $(0, 0)$ to $(n-1, m-1)$ in minimum time.

Since 2 or more paths can lead to same cell with different time → **DJKSTRA'S**

twist →

first move → 1 second

second move → 2 second

third move → 1 second

∴ (alternate (1, 2, 1))

```
long long newdiss = currdiss + movecost;
```

```
if (currdiss < movetime[nr][nc]) {
```

```
    newdiss = movetime[nr][nc] + movecost; // laane me bhi time lagega
```

```
}
```

ho sakta he abhi tak room khula na ho

to wahi khada reh kar wait kardenge

```
if (state == 0) → movecost = 1
```

```
if (state == 1) → movecost = 2
```

C++ Implementation

```
class Solution {
public:
    int minTimeToReach(vector<vector<int>>& moveTime) {

        int n=moveTime.size();
        int m=moveTime[0].size();

        vector<pair<int,int>> dir={{-1,0},{0,-1},{1,0},{0,1}};

        priority_queue<
            pair<long long,pair<pair<int,int>,int>>,
            vector<pair<long long,pair<pair<int,int>,int>>>,
            greater<pair<long long,pair<pair<int,int>,int>>>> pq; //{time,
{{r,c},state}} , ek state rkhnna padega so that we know kab cost=1 and kab cost=2

        vector<vector<vector<long long>>> dist(n,vector<vector<long
long>>>(m,vector<long long>(2,LLONG_MAX)));
        pq.push({0,{0,0},0});
        dist[0][0][0]=0;

        while(!pq.empty()){
            int r=pq.top().second.first.first;
            int c=pq.top().second.first.second;
            long long currdiss=pq.top().first;
            int state=pq.top().second.second;

            pq.pop();
            if(currdiss>dist[r][c][state]) continue;
            if(r==n-1 && c==m-1) return currdiss;

            for(auto &it:dir){

                int nr=r+it.first;
                int nc=c+it.second;
                if(nr<0 && nc<0 && nr>n && nc>m) continue;
                if(state==0) cost=1;
                else if(state==1) cost=2;
                long long newdiss=currdiss+cost;

                if(currdiss//we are using currdiss , kyunki ho skta pohochne se pehle gate band
newdiss=moveTime[nr][nc]+cost; //we will wait jab tak
room khul na jaaye

                //aur phir 1 second to
            }
        }
    }
};
```

```
move in  
    }  
  
    int newstate=!state;  
  
    if(newdissreturn -1;  
  
    }  
};
```